



# CÉLIAN DI GIOVANNI

## PROFILE

Final Year Engineering Student (M.Sc. Level) specializing in Artificial Intelligence and Signal Processing. Experienced in **physiological signal analysis, feature extraction**, training and evaluation of **Machine Learning models** (Random Forest, SVM) on real data. Applied projects include **biometric classification, ECG signal analysis**, and **deployment of embedded AI models**.

Seeking a 6-month final-year internship starting April 1st, 2026.

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Digital and Electronic Engineering (M.Sc.) – eHealth Option</b><br>ISEN Méditerranée, Toulon                                       | 2023 - Present        |
| <b>Bionics Engineering – Erasmus Semester</b><br>Università di Pisa                                                                   | Sep. 2024 - Feb. 2025 |
| <b>CPGE PCSI / PC</b><br>Lycée International de Valbonne                                                                              | 2021 - 2023           |
| <b>French Scientific Baccalaureate (Mathematics &amp; Physics-Chemistry) - With Honors</b><br>Institution du Mont Saint-Jean, Antibes | 2021                  |

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |
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| <b>1st Prize at I-Novgames 2026 - Reinventing Patient Monitoring via Embedded Technology</b><br>Carelink Project - Anticipation and Detection of Epileptic Seizures <ul style="list-style-type: none"><li>Analyzed <b>ECG</b> signals from the <b>PhysioNet (CHB-MIT)</b> database.</li><li>Computed heart rate variability indicators: <b>RMSSD, SDNN, LF/HF</b>.</li><li>Trained <b>TensorFlow Lite AI</b> models using <b>Edge Impulse</b>.</li><li>Converted and compressed models for execution on STM32 targets.</li><li>Processed data from <b>10 patients and ~1000 30-second windows</b>.</li></ul> | 2026 |
| <b>Exploratory Results:</b> Detected ECG alterations consistent with documented seizure signatures. <b>Global Accuracy: 92%.</b><br>Functional Proof of Concept for a real-time anticipation and alert system.                                                                                                                                                                                                                                                                                                                                                                                               |      |
| <b>Skin Maven Bandage</b><br>Biometric Classification via AI (Master 1 Project) <ul style="list-style-type: none"><li>Built a database using skin impedance measurements collected from <b>83 volunteers</b>.</li><li>Applied data augmentation using <b>SMOTE</b> and randomization.</li><li>Trained and compared multiple models: Logistic Regression, Random Forest, Gaussian, KNN, SVC.</li></ul>                                                                                                                                                                                                        | 2025 |

**Results: Random Forest: 90% accuracy for gender, 60% for age, 20% for skin tone.**

## INFORMATION

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## SKILLS

- Python ●●●●●
- AI & Data Science: Classification, Deep Learning (CNN), Feature Engineering, Embedded AI ●●●●●
- Signal Processing: ECG, FFT, Bioimpedance ●●●●●
- Computer Vision: Segmentation, Otsu's Thresholding, Sobel, Mathematical Morphology ●●●●●
- Biostatistics ●●●●●
- Neurosciences ●●●●●
- Java / C ●●●●●

## LANGUAGES

- English - B2 Cambridge
- Spanish

INTERESTS

- Culture: Travel, History.
- Music & Arts: Acoustic and Electric Guitar, Cinema, Rock Music.
- Sports: Paddleboarding, Judo, Badminton, Running.
- Volunteering: Order of Malta Outreach (Social Patrols).

CERTIFICATES

- First Aid Certification (Level 1)

Region of Interest (ROI) Segmentation on X-ray Images

2025

- Full implementation in **Python and Octave**.
- Methods applied: Otsu’s thresholding, RGB/Grayscale conversion, erosion, dilation.
- Data processed: Brain and Chest X-rays (JPG format).

**Result: Functional and validated segmentation algorithm.**

PROFESSIONAL EXPERIENCE

Engineering Intern - QA & Clinical Data

June 2025 - Sept 2025

Nice University Hospital (CHU) - Pharmacy Department

- Performed functional validation of a medical monitoring mobile application (MUSE project).
- Defined **use cases and test scenarios**, and reported bugs/feedback to development teams.
- Acted as a liaison between medical staff and developers.
- Extracted and structured clinical data from unstructured **reports (DOCX/XLSX)**.
- Developed Python automation scripts (using openpyxl, python-docx).
- Centralized and analyzed data using **Excel / Power Query / Pivot Tables**.